

1. Design System & Visual Language

To ensure high performance, visual coherence, and professional clinical usability, the e-health.albania Health Center Management System (HCMS) utilizes a dedicated, highly rigid visual language designed explicitly for heavy data densities typical in hospital management environments.

1.1 Color Architecture

The layout applies a high-contrast palette carefully designed to prevent visual fatigue over long operational clinical shifts:

Primary Structural (#0F172A) Deep Navy for frames, sidebars, and structural headers.	Workspace Surface (#F8FAFC) Clinical Off-White to minimize screen glare and eye strain.	Action Accent (#0EA5E9) Medical Teal for interactive highlights and focus fields.	Critical Triage (#EF4444) Crimson Red reserved strictly for life-critical alerts.
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1.2 Typography & Scaling

The standard system rendering typography uses enterprise sans-serif families (**Inter** or **Segoe UI**) dynamically optimized across the application stack:

- **Main Application Headings:** 18pt–24pt Bold for core workspace identifiers.
- **Form Labels & Control Anchors:** 11pt Regular for descriptive fields.
- **Clinical Tabular Data:** 10pt Regular with strict fixed cell boundaries to guarantee alphanumeric strings are never clipped or obfuscated.

2. Typography, Controls, & Components

All reusable visual elements are wrapped inside encapsulated JavaFX control classes extending the base JavaFX layout nodes. This structure guarantees that design modifications apply application-wide without breaking backend operational models.

Component	JavaFX Structural Mapping	UX Design Behavior & Rules
Primary Navigation Sidebar	<code>javafx.scene.layout.VBox</code> bound inside a <code>BorderPane Left</code> area.	Includes collapsible text nodes, high-contrast flat vector icon states, and active role-permission visibility filtering.
Patient Records Grid	<code>javafx.scene.control.TableView</code> with custom cell factories.	Supports immediate sorting, asynchronous cell rendering, pagination, and inline status styling tags.
Clinical Form Layouts	<code>javafx.scene.layout.GridPane</code> with programmatic spacing constraints.	Strict vertical alignments. Highlights errors immediately using red border highlights and dynamic alert symbols.
Status Context Footer	<code>javafx.scene.layout.HBox</code> with spatial region distributions.	Displays constant database connections, system health stats, active background thread jobs, and live language tokens.

3. Language & Localization Strategy (NFR-Localisation)

The system layout features complete real-time localization between **English (EN)** and **Albanian (SQ - Shqip)** using `Java ResourceBundle` properties mapped natively into `JavaFX` properties. When the global language toggle is activated, all controls refresh their language bindings immediately without requiring an application restart or closing active patient worksheets.